Moran process

Three individuals, two dice, one survivor

Set up
Work out the fitnesses by hand.

Roll
One birth, one death, every step.

Pool
Count how often the hawk fixes.

Pairs playing:

Starting state: one hawk, two doves

	Hawk	Dove
Hawk	0	3
Dove	1	2

Four resources to share. Two hawks get ____ each; a hawk meeting a dove takes ____ to the dove's ____; two doves split them ____ each.



The number of hawks. Along the top a hawk is born; along the bottom one dies. The two ends are absorbing, so the process stops only once one type is gone.

A type's fitness counts what it scores against everyone *else* in the population, so it never plays itself.

State	f(hawk)	f(dove)
1 hawk, 2 doves		
2 hawks, 1 dove		

Which type is fitter when hawks are rare? And when they are common?

Birth is proportional to fitness. Death is uniform, so it is proportional to _____.

State	Type	Chosen for birth	Chosen for death
1 hawk, 2 doves	hawk		
1 hawk, 2 doves	dove		
2 hawks, 1 dove	hawk		
2 hawks, 1 dove	dove		

These are the numbers the dice have to reproduce, which is why we need a four and a six.

The number of hawks changes only when the type born differs from the type that dies.

$p_{10} =$ $p_{12} =$ $p_{21} =$ $p_{23} =$

Go back and write these on the chain. What happens with the remaining probability?

Runs that ended with all hawks: ____ out of ____, so a fixation probability of about ____.

The theory says $x_1 = \dfrac{1}{1 + \gamma_1 + \gamma_1\gamma_2}$ with $\gamma_i =$ ____, giving ____.

Close enough? ____ How many runs would we need to be sure? ____

What we just did in the room is **Question 1** on the Moran Process page, with a population of four instead of three.

(a)	The two random choices made at each step.	[3]
(b)	Fitnesses with two hawks and two doves.	[6]
(c)	The probability the number of hawks increases by one.	[7]
(d)	Define fixation, and say why there are exactly two absorbing states.	[5]
(e)	The probability it decreases, the ratio γ, and which way it drifts.	[4]

vknight.org/gt/topics/moran-process.html